

Chapter 5: Bases

In this part of the lecture notes we will introduce the notions of *linear dependence* and *linear independence*. Vectors $v_1, v_2, \dots, v_n$ are linearly independent if and only if the zero vector can be written in only one way as a linear combination of the v_j (namely as the trivial linear combination with 0-coefficients).

We will show that vectors v_j are linearly dependent (that is, are *not* linearly independent) if and only if one of the v_i is a linear combination of the other vectors v_j , that is, if and only if we can remove one of the v_i from the collection without changing the span.

Thus we can consider lists of vectors with as few elements as possible which span a given vector space V . Such “minimal” lists are automatically linearly independent. These considerations will lead into the theory of bases and dimension to be developed later.

A basis for a vector space V is a collection of *linearly independent* vectors that generates V . It is a shortest-possible list of vectors generating V , in the sense that removing any of the vectors results in a collection that does *not* span V .

Bases can also be characterised by the property that any vector in V can be written as a linear combination of the basis vectors *in a unique way*.

Bases are of great theoretical and practical importance in mathematics. Knowing a basis is tantamount to knowing the vector space.

We will soon see that *every finitely generated vector space has a basis*. But first we need to define the notion of a basis, and study its fundamental properties.

Section 5.1: Definition and examples

1 Basis of a vector space

Definition. Let V be an $\mathbb{R}$ -vector space.

We call the vectors $v_1, v_2, \dots, v_n \in V$ a *basis for* V if the vectors generate V , and if they are linearly independent.

That is, to form a basis the vectors $v_1, v_2, \dots, v_n$ need to satisfy the following two conditions:

- $V = \text{span} \{v_1, v_2, \dots, v_n\}$
- For all given scalars $\lambda_i \in \mathbb{R}$, whenever $\sum_{j=1}^n \lambda_j v_j = \underline{0}$ then necessarily $\lambda_j = 0$ for all j .

2 The unit vectors in $\mathbb{R}^n$ form a basis

The unit vectors $e_1, e_2, \dots, e_n \in \mathbb{R}^n$ form a basis of $\mathbb{R}^n$.

Proof. We have established all the relevant facts before:

- On slide 9 of Chapter 4 we proved **linear independence**.
- From slides 10–11 of Chapter 3 we know that $\text{span} \{e_1, \dots, e_n\} = \mathbb{R}^n$, that is, the vectors $e_1, e_2, \dots, e_n$ **generate** $\mathbb{R}^n$. — Of course we can spell it out explicitly: the equality

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$$

says that every element of $\mathbb{R}^n$ is a linear combination of the unit vectors. ■

3 Monomials in $\mathcal{P}_n$ form a basis

The $n+1$ monomials $x^n, x^{n-1}, \dots, x^2, x, 1$ form a basis of the $\mathbb{R}$ -vector space $\mathcal{P}_n$.

In fact, by definition of "polynomial of degree at most n ", every element of $\mathcal{P}_n$ is a linear combination of these monomials! Thus

$$\mathcal{P}_n = \text{span} \{x^n, x^{n-1}, \dots, x^2, x, 1\} .$$

Linear independence is also easy:

$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is the zero polynomial, by definition, only if all the coefficients a_j are zero. ■

4 A basis for $\mathcal{M}_{m,n}$

Let $E_{i,j}$ denote the matrix with m rows and n columns which has entry 1 in row i and column j , and all other entries 0.

The matrices $E_{i,j}$ with $1 \leq i \leq m$ and $1 \leq j \leq n$ form a basis for $\mathcal{M}_{m,n}$.

This is analogous to the case of unit vectors in $\mathbb{R}^n$.

Explicitly, the following 6 matrices form a basis for $\mathcal{M}_{2,3}$:

$$E_{1,1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad E_{1,2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad E_{1,3} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$E_{2,1} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad E_{2,2} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad E_{2,3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

5 A basis for $\mathcal{P}_2$

In the $\mathbb{R}$ -vector space $\mathcal{P}_2$, the vectors

$$p_1 = x^2 + x + 1, \quad p_2 = x^2 + x + 2, \quad p_3 = x^2 + 2x + 2$$

are linearly independent and generate $\mathcal{P}_2$, hence form a basis.

Let's verify the claim:

- Linear independence was established on slide 8 of Chapter 4.
- Need to check that $\text{span}\{p_1, p_2, p_3\} = \mathcal{P}_2$.

That is, given any $p = a_2x^2 + a_1x + a_0 \in \mathcal{P}_2$, can we find scalars $\lambda_1, \lambda_2, \lambda_3$ with $p = \lambda_1p_1 + \lambda_2p_2 + \lambda_3p_3$?

The resulting system of linear equations always has a solution:

$$\lambda_1 = 2a_2 - a_0 \quad \lambda_2 = a_0 - a_1 \quad \lambda_3 = a_1 - a_2 .$$

So p_1, p_2, p_3 generate $\mathcal{P}_2$ as desired. ■

6 Unit sequences in $\mathcal{S}$

In the vector space $\mathcal{S}$ of real sequences, denote by

$$b(n) = (b(n)_1, b(n)_2, \dots)$$

the sequence which is zero everywhere except at position n where it has entry 1, that is,

$$b(n)_n = 1 \quad \text{and} \quad b(n)_k = 0 \text{ for all } k \neq n .$$

Explicitly,

$$b(1) = (1, 0, 0, 0, \dots, 0, \dots)$$

$$b(2) = (0, 1, 0, 0, \dots, 0, \dots)$$

$$b(3) = (0, 0, 1, 0, \dots, 0, \dots)$$

and so on. These are the "*unit sequences*", in analogy to the "unit vectors" we have met earlier.

7 The unit sequences do not form a basis for $\mathcal{S}$

It is quite easy to see that the unit sequences $b(n)$, $n = 1, 2, \dots$ are linearly independent. For example, suppose we have $\mu_1 b(1) + \mu_2 b(2) + \mu_3 b(3) = \underline{0}$. Then just looking at the first term of the sequences we have

$$0 = \mu_1 b(1)_1 + \mu_2 b(2)_1 + \mu_3 b(3)_1 = \mu_1$$

(since $b(n)_1 = 0$ for $n \neq 1$ and $b(1)_1 = 1$). Looking at the second and third term yields $\mu_2 = 0$ and $\mu_3 = 0$ as well.

However, the unit sequences $b(n)$, $n = 1, 2, \dots$ do **not** generate the vector space $\mathcal{S}$. Every element s in

$$U = \text{span} \{b(1), b(2), b(3), \dots\}$$

is a **finite** sum of sequences of the form $\lambda b(n)$, so s can have only finitely many non-zero terms! For example, the constant sequence $(\pi, \pi, \pi, \dots)$ is an element of $\mathcal{S}$ but not of U .

8 The unit sequences form a basis for $\mathcal{S}_{\text{fin}}$

Let $\mathcal{S}_{\text{fin}}$ denote the subspace of $\mathcal{S}$ consisting of sequences with finitely many non-zero terms. This is sometimes called the set of "finite sequences".

The unit sequences form a basis for $\mathcal{S}_{\text{fin}}$.

We have already explained that $U = \text{span} \{b(1), b(2), b(3), \dots\}$ is a subset of $\mathcal{S}_{\text{fin}}$. Conversely, if $s = (s_n)_{n \in \mathbb{N}}$ is a finite sequence then there is a number N with $s_k = 0$ for $k > N$. Then

$$s = s_1 b(1) + s_2 b(2) + \dots + s_N b(N)$$

is an element of U . — This shows $U = \mathcal{S}_{\text{fin}}$. The unit sequences are linearly independent and generate U (by definition of U), hence form a basis. ■

Section 5.2: Characterising bases

By definition, a basis is a collection of linearly independent vectors that generate the vector space. There are other important properties: a basis

- has the property that *every* vector of the vector space can be expressed as a linear combination *in a unique way*;
- is a *minimal* list of vectors generating the vector space;
- is a *maximal* list of linearly independent vectors.

These properties are characteristic for a basis: If a collection of vectors satisfies one of these, the vectors actually form a basis (and hence satisfy all these properties).

9 Every vector can be written as a linear combination of basis vectors in a unique way

Let V an $\mathbb{R}$ -vector space, and let $v_1, v_2, \dots, v_n$ be given elements of V . The following statements are equivalent:

- (i) $v_1, v_2, \dots, v_n$ form a basis for V
- (ii) **every** vector in V **can** be expressed **in a unique way** as a linear combination of the vectors $v_1, v_2, \dots, v_n$

Statement (ii) actually says two things:

- Every vector *can* be written as a linear combination of the v_j : given $v \in V$ there exist scalars λ_j with $v = \sum_{j=1}^n \lambda_j v_j$.
- The "uniqueness" part means that if $\sum_{j=1}^n \lambda_j v_j = \sum_{j=1}^n \mu_j v_j$ are "two" ways of writing the same vector, then $\lambda_j = \mu_j$ for all j .

10 Proof of (i) $\Rightarrow$ (ii)

Suppose that $v_1, v_2, \dots, v_n$ form a basis. Then we know $V = \text{span}\{v_1, v_2, \dots, v_n\}$, so every vector in V is a linear combination of the v_j : for any given $v \in V$ we find scalars $\lambda_j \in \mathbb{R}$ such that

$$v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n .$$

We need to show that this is the **only** way of writing v .

So suppose that we have another representation,

$$v = \mu_1 v_1 + \mu_2 v_2 + \dots + \mu_n v_n ,$$

with scalars μ_j . By subtracting this implies

$$(\lambda_1 - \mu_1)v_1 + (\lambda_2 - \mu_2)v_2 + \dots + (\lambda_n - \mu_n)v_n = v - v = \underline{0} ,$$

and **since $v_1, \dots, v_n$ are linearly independent** we must have $\lambda_j - \mu_j = 0$ for all j , that is, $\lambda_j = \mu_j$, for each $j = 1, 2, \dots, n$.

In other words, the second representation of v is the same as the first: the representation is unique.

11 Proof of (ii) $\Rightarrow$ (i)

Conversely, suppose that **every** vector in V **can** be expressed **in a unique way** as a linear combination of $v_1, v_2, \dots, v_n$.

- "Can be expressed" means that every vector of V is in $\text{span}\{v_1, v_2, \dots, v_n\}$ so that $V = \text{span}\{v_1, v_2, \dots, v_n\}$.

That is, the vectors $v_1, v_2, \dots, v_n$ generate V .

- Next, suppose that $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \underline{0}$, for scalars $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$.

We certainly also have the equality

$$0v_1 + 0v_2 + \dots + 0v_n = \underline{0} .$$

But by hypothesis, the zero vector can be written in a **unique** way as a linear combination of $v_1, v_2, \dots, v_n$. So we **must** have $\lambda_i = 0$ for $1 \leq i \leq n$. That is, the vectors $v_1, v_2, \dots, v_n$ are linearly independent. ■

12 Application: Change of coordinates

What geometric object does the equation

$$\frac{7}{4}x^2 - \frac{\sqrt{3}}{2}xy + \frac{13}{4}y^2 = 1$$

describe?

We think of a solution x, y as the point with x-coordinate x and y-coordinate y . Of course

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = xe_1 + ye_2$$

is a linear combination of the unit vectors e_1 and e_2 , which form a basis for $\mathbb{R}^2$.

Let us try to use a different basis for $\mathbb{R}^2$, given by

$$w_1 = \begin{pmatrix} \sqrt{3}/4 \\ -1/4 \end{pmatrix} \quad \text{and} \quad w_2 = \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix}.$$

13 Application: Change of coordinates (continued)

The "point" $\begin{pmatrix} x \\ y \end{pmatrix}$ is also a linear combination of w_1 and w_2 ,

$$\begin{pmatrix} x \\ y \end{pmatrix} = aw_1 + bw_2 = a \begin{pmatrix} \sqrt{3}/4 \\ -1/4 \end{pmatrix} + b \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix},$$

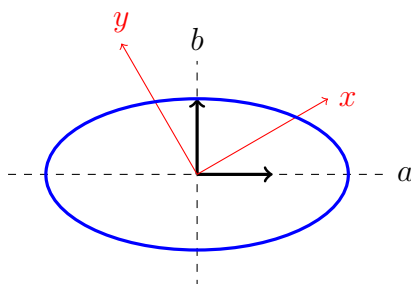
with uniquely determined coefficients a and b . Note that x, y, a and b are linked by a system of linear equations:

$$\left. \begin{array}{l} \frac{\sqrt{3}}{4}a + \frac{1}{4}b = x \\ -\frac{1}{4}a + \frac{\sqrt{3}}{4}b = y \end{array} \right\}$$

Putting these expressions for x and y into the original equation and simplifying gives the equation $a^2/4 + b^2 = 1$. So in a, b -coordinates we have an ellipse! The original equation describes a rotated and dilated version of this ellipse in x, y -coordinates.

14 Application: Change of coordinates (picture in a,b -coordinates)

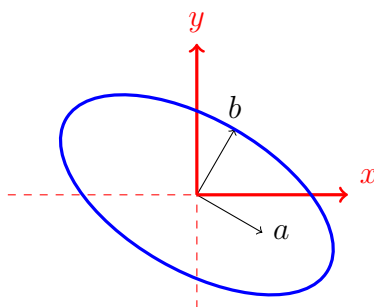
Here is a picture of the ellipse in a,b -coordinates; the black arrows indicate the vectors w_1 (in a -direction) and w_2 (in b -direction), the red arrows indicate the vectors e_1 and e_2 as expressed in a,b -coordinates.



In **blue**: solutions of the equation $\frac{1}{4}a^2 + b^2 = 1$

15 Application: Change of coordinates (picture in x,y -coordinates)

And here is a picture of the ellipse in x,y -coordinates; the red arrows indicate the vectors e_1 (in x -direction) and e_2 (in y -direction), the black arrows indicate the vectors w_1 and w_2 as expressed in x,y -coordinates.



In **blue**: solutions of the equation $\frac{7}{4}x^2 - \frac{\sqrt{3}}{2}xy + \frac{13}{4}y^2 = 1$

Characterising bases as extremal sets

In short, the characterisation of a basis is:

- Add a vector – loose linear independence
- Delete a vector – loose the "generation" property

This makes a basis an "extremal set" in the sense that you can't make it bigger or smaller without losing the "basis" property.

16 Bases are extremal sets

Let V be an $\mathbb{R}$ -vector space, and let $v_1, v_2, \dots, v_n$ be vectors in V .

The following conditions are equivalent:

- (i) The vectors $v_1, v_2, \dots, v_n$ generate the vector space V and are linearly independent (that is, they form a basis for V).
- (ii) The vectors $v_1, v_2, \dots, v_n$ form a minimal list of vectors generating V : they generate V , but no $n - 1$ of these vectors generate V .
- (iii) The vectors $v_1, v_2, \dots, v_n$ are a maximal list of linearly independent vectors: $v_1, v_2, \dots, v_n$ are linearly independent, but given any vector $w \in V$, the vectors $w, v_1, v_2, \dots, v_n$ are linearly dependent.

If these conditions are satisfied, V is a finitely generated vector space with basis $v_1, v_2, \dots, v_n$.

17 Bases are extremal sets: proof of (i) $\Leftrightarrow$ (ii)

Suppose (i) holds: $v_1, v_2, \dots, v_n$ form a basis. Then these vectors generate V . They are also linearly independent. Hence from slide 22 in Chapter 4 we know that for any k ,

$$\text{span}\{v_1, v_2, \dots, v_{k-1}, v_{k+1}, \dots, v_n\} \neq \text{span}\{v_1, v_2, \dots, v_n\} = V.$$

So no $n - 1$ of the vectors generate V . This means that (ii) holds.

Conversely, suppose that (ii) holds. Then $v_1, v_2, \dots, v_n$ generate V , and the minimality condition means that

$$\text{span}\{v_1, v_2, \dots, v_{k-1}, v_{k+1}, \dots, v_n\} \neq \text{span}\{v_1, v_2, \dots, v_n\} = V$$

for any k . Thus by slide 21 of Chapter 4 the vectors are linearly independent. This verifies statement (i).

18 Bases are extremal sets: proof of (i) $\Leftrightarrow$ (iii)

Suppose (i) holds: $v_1, v_2, \dots, v_n$ form a basis. Then these vectors are linearly independent. They also generate V so for any $w \in V$,

$$w \in \text{span}\{v_1, v_2, \dots, v_n\},$$

that is, the vectors $w, v_1, v_2, \dots, v_n$ are linearly dependent (slide 15 of Chapter 4). So (iii) holds.

Conversely, suppose (iii) holds. Then $v_1, v_2, \dots, v_n$ are linearly independent. Given $w \in V$ we know that $w, v_1, v_2, \dots, v_n$ are linearly dependent, by "maximality". Hence there are scalars $\lambda, \lambda_1, \dots, \lambda_n$, not all zero, with

$$\lambda w + \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \underline{0}. \quad (\dagger)$$

We must have $\lambda \neq 0$; else one of the λ_i is non-zero, and $(\dagger)$ says that the v_i are linearly dependent, contradicting (iii). — So we get

$$w = -\frac{\lambda_1}{\lambda}v_1 - \frac{\lambda_2}{\lambda}v_2 - \dots - \frac{\lambda_n}{\lambda}v_n,$$

that is, $w \in \text{span}\{v_1, \dots, v_n\}$: the vectors v_i generate V . ■

19 Summary

- Vectors $v_1, v_2, \dots, v_n$ form a basis for V if they are linearly independent and generate V .
- Example: $e_1, e_2, \dots, e_n \in \mathbb{R}^n$ form a basis for $\mathbb{R}^n$.
- Example: $x^n, x^{n-1}, \dots, x, 1$ form a basis for $\mathcal{P}_n$.

- A basis can be characterised as a minimal generating list of vectors, and also as a maximal linearly independent list of vectors.
- A basis allows to express *every* vector in the vector space as a linear combination of the basis vectors *in a unique way*. This property is characteristic for a basis.